

Database Schema Documentation

System: Pizza Sales Management System | Document Type: Schema & DDL Specification

Overview: This document specifies the relational database schema design derived from the ER Diagram for the Pizza Sales System. It contains the detailed table dictionary, entity relationships, and execution-ready SQL DDL code.

1. Data Dictionary / Relational Schema

1. Table: `pizza_types`

Stores master data for pizza categories, names, and ingredient lists.

Column Name	Data Type	Key / Constraint	Description
<code>pizza_type_id</code>	<code>VARCHAR(50)</code>	PRIMARY KEY	Unique identifier for the pizza type
<code>name</code>	<code>VARCHAR(100)</code>	NOT NULL	Name of the pizza
<code>category</code>	<code>VARCHAR(50)</code>	NOT NULL	Category (e.g., Classic, Veggie, Supreme)
<code>ingredients</code>	<code>TEXT</code>	NULLABLE	Comma-separated list of ingredients

2. Table: `pizzas`

Stores specific pizza variants (size and pricing) linked to a pizza type.

Column Name	Data Type	Key / Constraint	Description
<code>pizza_id</code>	<code>VARCHAR(50)</code>	PRIMARY KEY	Unique identifier for specific pizza variant
<code>pizza_type_id</code>	<code>VARCHAR(50)</code>	FOREIGN KEY	References <code>pizza_types</code> (<code>pizza_type_id</code>)
<code>size</code>	<code>VARCHAR(10)</code>	NOT NULL	Size code (S, M, L, XL, XXL)
<code>price</code>	<code>DECIMAL(10,2)</code>	NOT NULL	Price of the pizza variant

3. Table: `orders`

Stores order transaction headers.

Column Name	Data Type	Key / Constraint	Description
<code>order_id</code>	<code>INT</code>	PRIMARY KEY	Unique identifier for the order

date	DATE	NOT NULL	Date when the order was placed
time	TIME	NOT NULL	Time when the order was placed

4. Table: order_details

Junction table storing line items for each order.

Column Name	Data Type	Key / Constraint	Description
order_details_id	INT	PRIMARY KEY	Unique identifier for the line item
order_id	INT	FOREIGN KEY	References orders (order_id)
pizza_id	VARCHAR(50)	FOREIGN KEY	References pizzas (pizza_id)
quantity	INT	NOT NULL	Quantity of pizzas ordered

2. Entity Relationships

- **pizza_types (1) to pizzas (M)**: One pizza type can have multiple size variants.
- **orders (1) to order_details (M)**: One order can contain multiple detail item rows.
- **pizzas (1) to order_details (M)**: A specific pizza variant can appear in multiple order line items.

3. SQL Creation Script (DDL)

```
-- 1. Create Pizza Types Table
CREATE TABLE pizza_types (
    pizza_type_id VARCHAR(50) PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    category VARCHAR(50) NOT NULL,
    ingredients TEXT
);

-- 2. Create Pizzas Table
CREATE TABLE pizzas (
    pizza_id VARCHAR(50) PRIMARY KEY,
    pizza_type_id VARCHAR(50) NOT NULL,
    size VARCHAR(10) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
```

```
FOREIGN KEY (pizza_type_id) REFERENCES pizza_types(pizza_type_id)
);

-- 3. Create Orders Table
CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    date DATE NOT NULL,
    time TIME NOT NULL
);

-- 4. Create Order Details Table
CREATE TABLE order_details (
    order_details_id INT PRIMARY KEY,
    order_id INT NOT NULL,
    pizza_id VARCHAR(50) NOT NULL,
    quantity INT NOT NULL,
    FOREIGN KEY (order_id) REFERENCES orders(order_id),
    FOREIGN KEY (pizza_id) REFERENCES pizzas(pizza_id)
);
```